Nowhere Analyticity of Every Positive Compositional Iterate of the Fabius Function

A dyadic-spine proof through Faà di Bruno partitions

Research report prepared for the ProveIt Fabius-function corpus

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Result status. The theorem proved here is a new mathematical extension of the repository's formally verified base case. The proof is self-contained and has undergone a repository proof review; the delivered numerical diagnostic has also been reproduced exactly in its recovered Linux environment. Neither check is a Lean proof. Since the report snapshot, the finite block-size arithmetic of the partition defect has been formalized in Lean, but the finite-set-partition wrapper, composition theorem, weighted partition estimate, and spine asymptotic remain manuscript results.

Abstract

Let $F\colon [0,1] \to [0,1]$ be the bounded Fabius function and let $F^{\circ n}$ denote its n-fold compositional iterate. We prove that, for every integer $n \geq 1$, the smooth function $F^{\circ n}$ is real analytic at no point of $[0,1]$. In fact we establish the stronger statement that, for each fixed n, the Taylor series of $F^{\circ n}$ has radius of convergence zero at every point of a co-countable dense subset of $(0,1)$.

The proof has three ingredients. First, the exact derivative atlas for the Fabius function gives

$$F^{(m)}(x) = Q_m \Theta(2^m x), \qquad Q_m = 2^{\binom{m+1}{2}},$$

where Θ is the bounded signed global extension whose signs are governed by the Thue–Morse sequence. Second, a set-partition form of the Faà di Bruno formula is split into two extremal chain-rule terms and a genuinely branched remainder. A new "dyadic defect"

$$D(\pi) = \binom{m}{2} - \binom{|\pi|}{2} - \sum_{B \in \pi} \binom{|B|}{2}$$

shows that the total branched remainder is exponentially smaller than the extremal terms after normalization by Q_m. Iterating this reduction yields a finite spine sum indexed by the orbit $x, F(x), \ldots, F^{\circ(n-1)}(x)$. Third, the orbit dynamics make the corresponding exponential spine weights strictly unimodal, apart from a finite tie set. Away from ties, binary digit transitions force the dominant Thue–Morse amplitude to recur. At a tie, exact quarter-point values produce the explicit subsequence $m = 6\ell + 4$: the earlier maximal spine vanishes and the later one has amplitude exactly $\mathrm{up}(1/9) \geq 1/2$. The resulting derivative lower bound grows like Q_m times an ordinary exponential, far faster than $m!$, and hence forces zero Taylor radius.

The report also records consequences for inverse iterates and local elementary representations, gives a reproducible sinc-product computation, proves the exact tie-point noncancellation mechanism, and formulates concrete research directions involving q-Bell polynomials, Lambert–W endpoint transseries, Legendre expansions, and formalization.

Contents

1 Statement of the problem and the strengthened result

The bounded Fabius function is the unique smooth increasing bijection $F\colon [0,1] \to [0,1]$ satisfying

$$F(0) = 0, \qquad F(1-x) = 1 - F(x), \qquad F'(x) = 2F(2x) \quad (0 \le x \le \tfrac{1}{2}), \tag{1}$$

with the usual constant continuation outside the unit interval when derivatives at endpoints are discussed. We write

$$G_n := F^{\circ n}, \qquad G_0 := \mathrm{id}\,.$$

Every G_n is C^∞ because it is a finite composition of smooth functions. The question is whether composition might suppress the derivative oscillation that makes F itself nowhere analytic. The answer is no.

Theorem 1.1 (Main theorem). *For every integer $n \ge 1$, the compositional iterate $G_n = F^{\circ n}$ is not real analytic at any point of $[0,1]$.*

The proof actually gives considerably more.

Theorem 1.2 (Co-countable zero-radius set). *Fix $n \ge 1$. There exists a co-countable dense set $E_n \subset (0,1)$ such that, for every $x \in E_n$,*

$$\limsup_{m\to\infty} \left(\frac{\left|G_n^{(m)}(x)\right|}{m!} \right)^{1/m} = +\infty. \tag{2}$$

Consequently the formal Taylor series of G_n at every $x \in E_n$ has radius of convergence zero.

The distinction between theorem 1.2 and theorem 1.1 is useful. At a small exceptional countable set, the signed extension can vanish eventually at a dyadic orbit point, and several orbit spines can tie. The tie itself is no longer an obstruction: proposition 9.3 proves zero Taylor radius at every tie point. A complete classification of the remaining dyadic-orbit jets is unnecessary for the main theorem: the zero-radius set is dense, whereas analyticity at one point would propagate to a whole neighborhood.

2 Repository audit and novelty boundary

The source audit was organized around the repository's canonical documents and its own consolidation manifests. This is important because the tree contains live proof-backed expositions, a formal Lean development, canonical frontier syntheses, and many frozen or mechanically merged historical drafts. Treating every duplicate as an independent source would obscure rather than improve provenance.

2.1 Canonical sources inspected

The following live documents form the mathematical backbone of the present report.

Repository document	Role in the present proof
`Fabius_Function_and_Rvachev_Up.tex`	Primary proof-backed exposition: definitions, symmetry, strict monotonicity, convexity and concavity, the signed global extension, exact all-order derivative formulae, sharp derivative bounds, dyadic jets, inverse-function facts, and the formal base nowhere-analyticity statement.

Repository document	Role in the present proof
`Fabius_Function_Mathematical_Glossary.tex`	Notation and cross-reference audit, including the bounded function, signed global extension, Rvachev bump, dyadic grids, Thue–Morse signs, and analytic-locus language.
`Non_Elementarity_of_the_Fabius_Function.tex`	Dense analyticity of elementary-function classes, the exact analytic locus of F, and the corresponding obstruction for F^{-1}.
`fabius_lean_walkthrough.tex`	Module-level proof map for the formal derivative, flatness, inverse, and nowhere-analyticity results.
`semi-formalized-research-frontiers.tex`	Canonical synthesis of q-binomial, sinc-product, Lambert–W, Bell/Bernoulli, Legendre, inverse, sampling, and asymptotic directions.

The archive index and the semi-formalized-frontier README were used as provenance crosswalks for first versions, absorbed notebooks, mechanically merged draft groups, and editorial syntheses. Thus historical copies were audited through their canonical successors and checksummed consolidation records rather than cited as competing statements. The repository itself explicitly separates proof-backed claims from frontier statements awaiting literal formal counterparts; this report follows the same convention.

2.2 Existing formal result and the new step

The repository already proves the exact base analytic locus

$$F \text{ is analytic at } x \quad\Longleftrightarrow\quad x \notin [0,1]. \tag{3}$$

The formal proof for the Rvachev bump uses dense odd dyadic centers where all sufficiently high derivatives vanish, forcing a hypothetical local analytic expansion to be a polynomial and then contradicting a nearby nonzero derivative. Reflection transfers the obstruction to the bounded Fabius function.

No audited canonical source states or proves the corresponding theorem for every positive self-composition $F^{\circ n}$. The base proof does not automatically pass through composition: at a dyadic point of the outer or inner function, the full Faà di Bruno expansion contains many mixed terms, so eventual vanishing of a single jet is not preserved. The new contribution is therefore not a cosmetic induction. It is the composition-stable asymptotic separation developed in sections 5 and 7.

3 Fabius, Rvachev, Thue–Morse, and the derivative atlas

3.1 The Rvachev bump and the signed global extension

Let up denote Rvachev's compactly supported C^∞ function, normalized by

$$\operatorname{supp}\mathrm{up} = [-1,1], \qquad \mathrm{up}(0) = 1,$$

and related to the bounded Fabius function by

$$\mathrm{up}(t) = F(1-|t|) \qquad (-1 \le t \le 1). \tag{4}$$

In particular, up is even, strictly decreasing on $[0,1]$, and

$$\mathrm{up}(\tfrac12) = F(\tfrac12) = \tfrac12. \tag{5}$$

For $k \geq 0$, put

$$\varepsilon_k = (-1)^{s_2(k)}, \tag{6}$$

where $s_2(k)$ is the sum of the binary digits of k. The signed global extension Θ can be written blockwise as

$$\Theta(t) = \varepsilon_k \ \mathrm{up}\big(t - (2k+1)\big), \qquad 2k \leq t \leq 2k+2. \tag{7}$$

It agrees with F on $[0, 1]$, satisfies $\Theta'(t) = 2\Theta(2t)$ globally, and obeys

$$|\Theta(t)| \leq 1. \tag{8}$$

The translate signs in (7) are precisely the Thue–Morse signs.

3.2 Exact all-order derivatives

Define the superfactorial dyadic scale

$$Q_m := 2^{\binom{m+1}{2}} = 2^{m(m+1)/2}. \tag{9}$$

Repeated differentiation of the dilation equation gives, for every $m \geq 1$ and $x \in (0, 1)$,

$$F^{(m)}(x) = Q_m \Theta(2^m x). \tag{10}$$

Consequently

$$\left|F^{(m)}(x)\right| \leq Q_m. \tag{11}$$

At the matched dyadic grid this becomes the exact Thue–Morse jet law

$$F^{(m)}\left(\frac{a}{2^m}\right) = \begin{cases} 0, & a \text{ even}, \\ Q_m \varepsilon_{\lfloor a/2 \rfloor}, & a \text{ odd}. \end{cases} \tag{12}$$

The factor Q_m is the source of nonanalytic growth. Since

$$Q_m^{1/m} = 2^{(m+1)/2}, \tag{13}$$

no fixed ordinary exponential can neutralize it after division by $m!$. The only serious issue is cancellation among the many chain-rule terms created by composition.

3.3 Shape and orbit dynamics

The repository proves that F is a strictly increasing bijection of $[0, 1]$, that F' is strictly increasing on $[0, \frac{1}{2}]$ and strictly decreasing on $[\frac{1}{2}, 1]$, and that

$$F'(0) = 0, \qquad F'(\tfrac{1}{2}) = 2, \qquad F'(1) = 0. \tag{14}$$

Moreover F is strictly convex on $[0, \frac{1}{2}]$ and strictly concave on $[\frac{1}{2}, 1]$. Comparing with the chord joining $(0, 0)$ and $(\frac{1}{2}, \frac{1}{2})$ gives

$$F(x) < x \quad (0 < x < \tfrac{1}{2}), \qquad F(x) > x \quad (\tfrac{1}{2} < x < 1). \tag{15}$$

Thus every noncentral interior orbit moves monotonically toward an endpoint:

$$0 < x < \tfrac{1}{2} \implies x > F(x) > F^{\circ 2}(x) > \cdots > 0, \tag{16}$$

$$\tfrac{1}{2} < x < 1 \implies x < F(x) < F^{\circ 2}(x) < \cdots < 1. \tag{17}$$

This elementary dynamical fact will force strict unimodality of the exponential weights in the derivative spine sum.

4 Why a direct dyadic-jet induction fails

It is useful to see the obstruction before introducing the cure. For two smooth functions f and h,

$$(f \circ h)^{(m)}(x) = \sum_{\pi \in \Pi_m} f^{(|\pi|)}(h(x)) \prod_{B \in \pi} h^{(|B|)}(x), \tag{18}$$

where Π_m is the set of partitions of $\{1, \dots, m\}$. Even if $h^{(r)}(x) = 0$ for all large r, partitions with many small blocks can survive. Similarly, even if the high derivatives of f vanish at $h(x)$, the term $f'(h(x))h^{(m)}(x)$ survives. There is no pointwise “eventually zero jet” induction for $F^{\circ n}$.

At the opposite extreme, a crude absolute-value bound on (18) introduces Bell numbers. Since the number of partitions is superexponential on an ordinary exponential scale, such a bound loses the delicate comparison with Q_m. The correct normalization must exploit the quadratic exponent in Q_m, not merely count partitions. This leads to the defect below.

5 The dyadic defect of a Faà di Bruno partition

Let $\pi \in \Pi_m$ have $k = |\pi|$ blocks of sizes $r_1, \dots, r_k$, so $r_1 + \dots + r_k = m$. Define

$$D(\pi) := \binom{m}{2} - \binom{k}{2} - \sum_{i=1}^{k} \binom{r_i}{2}. \tag{19}$$

An equivalent and more transparent formula is

$$D(\pi) = \sum_{1 \le i < j \le k} (r_i r_j - 1). \tag{20}$$

Every summand is nonnegative. Therefore $D(\pi) = 0$ only in the two extremal cases: there is one block, or every block is a singleton. These are exactly the two unbranched chain-rule spines.

Let $u(\pi)$ be the number of singleton blocks and let

$$s(\pi) := k - u(\pi) \tag{21}$$

be the number of non-singleton blocks.

Lemma 5.1 (Quadratic-scale factorization)**.** *For every partition* $\pi \in \Pi_m$,

$$\frac{Q_k \prod_{r_i \ge 2} Q_{r_i}}{Q_m} = 2^{s(\pi) - D(\pi)}. \tag{22}$$

Proof. Using $\log_2 Q_r = r(r+1)/2$ and omitting singleton blocks from the product,

$$\begin{aligned} 2 \log_2 \frac{Q_k \prod_{r_i \ge 2} Q_{r_i}}{Q_m} &= k(k+1) + \sum_{r_i \ge 2} r_i(r_i+1) - m(m+1) \\ &= k^2 + s(\pi) + \sum_{r_i \ge 2} r_i^2 - m^2. \end{aligned}$$

On the other hand, expanding (19) gives

$$2D(\pi) = m^2 - k^2 + s(\pi) - \sum_{r_i \ge 2} r_i^2.$$

Subtracting proves (22). □

The next proposition is the combinatorial engine. Its role is stronger than saying that each nonextremal partition has positive defect: it shows that the sum over all of them is still exponentially small.

Proposition 5.2 (Weighted defect decay). *For each fixed $K > 0$ there exist constants $C_K > 0$ and $\rho_K \in (0,1)$ such that*

$$\sum_{\substack{\pi\in\Pi_m\\ 1<|\pi|<m}} (2K)^{s(\pi)} 2^{-D(\pi)} \le C_K \rho_K^m \qquad (m \ge 2). \tag{23}$$

Proof. We give the full counting argument because this is the point at which a naive Bell number estimate fails.

Write $k = |\pi|$, let

$$e := m - k \tag{24}$$

be the total block excess over singletons, and let $r_{\max} = m - t$ be the largest block size. Convexity of $r \mapsto \binom{r}{2}$ shows that, for fixed m and k, the defect is minimized by one block of size $m - k + 1$ and $k - 1$ singleton blocks. Hence

$$D(\pi) \ge e(k-1) = e(m-e-1). \tag{25}$$

The contribution between a largest block and all other blocks gives the second bound

$$D(\pi) \ge (m-t)t - (k-1) \ge t(m-t-1). \tag{26}$$

First consider partitions with $1 \le e \le m/4$. Their non-singleton blocks contain at most $2e$ elements, because the number of such blocks is at most e. Choosing those elements and partitioning them gives at most

$$(2e+1)m^{2e}(2e)^{2e} \tag{27}$$

possibilities. Also $s(\pi) \le e$, while (25) gives $D(\pi) \ge e(3m/4 - 1)$. Uniformly for $1 \le e \le m/4$, the base-two logarithm of the counting and weight factor is at most

$$\log_2(2e+1) + 4e\log_2 m + e\log_2 \max(1, 2K).$$

For fixed K this is at most $em/4$ once m is large enough, whereas the negative defect exponent is at most $-e(3m/4-1)$. Thus the e-th contribution is at most $2^{-em/3}$ after enlarging the threshold once more, uniformly in e. The resulting geometric sum is $\mathcal{O}(2^{-m/3})$.

Next consider partitions with $1 \le t \le m/4$. Choose the t elements outside a largest block and partition them. Up to harmless overcounting there are at most $m^t t^t$ choices, and $s(\pi) \le t+1$. By (26), $D(\pi) \ge t(3m/4 - 1)$. The same explicit estimate as above, now with t in place of e (and one harmless extra factor $\max(1, 2K)$), bounds the t-th contribution by $2^{-tm/3}$ for all sufficiently large m, uniformly for $1 \le t \le m/4$. Its sum is again exponentially small.

It remains to treat the middle region $e > m/4$ and $t > m/4$. If $t \le 3m/4$, then (26) gives $D(\pi) \ge c_1 m^2$ for all sufficiently large m and an absolute $c_1 > 0$. Suppose instead that $t > 3m/4$, so every block has size less than $m/4$. If $k > m/4$, then (25) gives $D(\pi) \ge c_2 m^2$. If $k \le m/4$, then

$$\begin{aligned} D(\pi) &= \sum_{i<j} r_i r_j - \binom{k}{2} \\ &= \frac{m^2 - \sum_i r_i^2}{2} - \binom{k}{2} > \frac{3m^2}{8} - \frac{m^2}{32} = \frac{11m^2}{32}. \end{aligned}$$

In the middle region, the total number of partitions is at most m^m and the weight $(2K)^{s(\pi)}$ is at most $\max(1, 2K)^m$. Both are overwhelmed by 2^{-cm^2}. Combining the three regions proves (23), after increasing C_K to cover the finitely many small values of m. □

6 A two-spine composition lemma

The weighted defect estimate converts Faà di Bruno's formula into an asymptotically binary chain rule.

Lemma 6.1 (Two-spine reduction). *Let h be smooth near x, let f be smooth near $y = h(x)$, and put $B = |h'(x)|$. Suppose there are constants $H > 0$ and $K \geq 1$ such that*

$$\left|f^{(r)}(y)\right| \leq Q_r \qquad (r \geq 1), \tag{28}$$

$$\left|h^{(r)}(x)\right| \leq K Q_r H^r \qquad (r \geq 2). \tag{29}$$

Then, with $L = \max(B, H)$,

$$(f \circ h)^{(m)}(x) = f'(y) h^{(m)}(x) + f^{(m)}(y) h'(x)^m + o(Q_m L^m) \quad (m \to \infty). \tag{30}$$

Proof. In (18), remove the one-block partition and the all-singleton partition. They give exactly the two displayed terms. For any remaining partition π with k blocks, u singleton blocks, and $s = k - u$ non-singleton blocks, the corresponding term has absolute value at most

$$Q_k B^u \prod_{r_i \geq 2} (K Q_{r_i} H^{r_i}) \leq Q_m L^m (2K)^s 2^{-D(\pi)},$$

where lemma 5.1 was used in the last step. Summing and applying proposition 5.2 bounds the entire remainder by $C_K Q_m L^m \rho_K^m = o(Q_m L^m)$. □

Remark 6.2 (Why this lemma is composition-stable). The estimate does not require analyticity, Gevrey bounds of factorial type, or a closed formula for every Bell polynomial. It uses only the exact quadratic dyadic scale Q_r and an ordinary exponential envelope H^r for the inner derivatives. The conclusion has the same form, so it can be iterated a fixed number of times.

7 The orbit-spine asymptotic for $F^{\circ n}$

Fix $x \in (0, 1)$ and $n \geq 1$. Define the finite orbit

$$x_j := G_j(x) = F^{\circ j}(x), \qquad 0 \leq j \leq n - 1, \tag{31}$$

and the local slopes

$$a_j := F'(x_j) > 0. \tag{32}$$

The prefix and suffix weights are

$$B_0 := 1, \qquad B_j := \prod_{r=0}^{j-1} a_r = G_j'(x) \quad (j \geq 1), \tag{33}$$

$$A_{j,n} := \prod_{r=j+1}^{n-1} a_r, \qquad 0 \leq j < n, \tag{34}$$

where empty products equal 1. Finally set

$$H_n(x) := \max_{0 \leq j < n} B_j. \tag{35}$$

Theorem 7.1 (Finite spine expansion). *For each fixed $n \geq 1$ and $x \in (0, 1)$,*

$$G_n^{(m)}(x) = Q_m \left[\sum_{j=0}^{n-1} A_{j,n} B_j^m \Theta(2^m x_j) + o\big(H_n(x)^m\big) \right] \qquad (m \to \infty). \tag{36}$$

Proof. We induct on n. For $n = 1$, $x_0 = x$, $A_{0,1} = B_0 = H_1 = 1$, and (36) is exactly (10).

Assume the formula for G_j. It immediately supplies a global-in-order envelope at the fixed point x: there is $K \geq 1$ such that

$$\left|G_j^{(r)}(x)\right| \leq K Q_r H_j(x)^r \qquad (r \geq 2). \tag{37}$$

Indeed the asymptotic gives the estimate for all sufficiently large r, and K can be enlarged to absorb the finitely many remaining orders.

Apply lemma 6.1 with $f = F$ and $h = G_j$. Since $h(x) = x_j$, $h'(x) = B_j$, $f'(x_j) = a_j$, and $f^{(m)}(x_j) = Q_m \Theta(2^m x_j)$, we obtain

$$G_{j+1}^{(m)}(x) = a_j G_j^{(m)}(x) + Q_m B_j^m \Theta(2^m x_j) + o\big(Q_m H_{j+1}(x)^m\big), \tag{38}$$

where $H_{j+1} = \max(H_j, B_j)$. Multiplication by a_j changes every old suffix factor $A_{i,j}$ into $A_{i,j+1}$, while the new term has suffix factor $A_{j,j+1} = 1$. This is precisely (36) for $j+1$. □

Remark 7.2 (Combinatorial interpretation). Each summand in (36) corresponds to a chain-rule tree with one high-valence vertex, located at orbit level j, and unary vertices everywhere else. All trees with two or more genuinely branching vertices are swallowed by the $o(H_n^m)$ remainder. This is why "spine" is an apt name.

8 Strict unimodality of the spine weights

The finite sum in (36) could still cancel if several B_j have the same maximal exponential size. The orbit dynamics show that this is exceptional.

Lemma 8.1 (Decreasing slope sequence). *Let $x \in (0,1) \setminus \{\frac{1}{2}\}$. Then the sequence*

$$a_0, a_1, a_2, \dots \tag{39}$$

is strictly decreasing.

Proof. If $x < \frac{1}{2}$, then (16) says $x_{j+1} < x_j < \frac{1}{2}$. Since F' is strictly increasing on the left half, $a_{j+1} = F'(x_{j+1}) < F'(x_j) = a_j$. If $x > \frac{1}{2}$, then $x_{j+1} > x_j > \frac{1}{2}$ and F' is strictly decreasing on the right half, giving the same conclusion. □

Lemma 8.2 (Unique or adjacent double maximum). *For $x \in (0,1) \setminus \{\frac{1}{2}\}$, the finite positive sequence $B_0, \dots, B_{n-1}$ is strictly log-concave in the ratio sense:*

$$\frac{B_{j+1}}{B_j} = a_j \quad \textit{is strictly decreasing in } j. \tag{40}$$

Hence its maximum is unique unless $a_k = 1$ for some $0 \leq k < n-1$; in that case the only maximizers are the adjacent pair $B_k = B_{k+1}$.

Proof. Equation (40) follows from (33) and lemma 8.1. A positive finite sequence whose successive ratios strictly decrease first rises while the ratio exceeds 1, possibly has one flat step when the ratio equals 1, and then falls. The conclusion follows. □

Because F' rises strictly from 0 to 2 on $[0, \frac{1}{2}]$, there is a unique $u \in (0, \frac{1}{2})$ with $F'(u) = 1$. Symmetry gives the only other solution $1 - u$. For a fixed n, the tie set is therefore

$$T_n := \bigcup_{\substack{k \geq 0 \\ k+1 < n}} G_k^{-1}\{u, 1-u\}. \tag{41}$$

Thus $T_1 = \varnothing$. Each G_k is a homeomorphism of $[0,1]$, so in general T_n is finite, with at most $2(n-1)$ points.

9 Binary transitions force recurrent large amplitudes

Let $\mathcal{D}_2$ denote the dyadic rationals in $(0,1)$.

Lemma 9.1 (Binary-transition lemma)**.** *If $y \in (0,1) \setminus \mathcal{D}_2$, then there are infinitely many $m \geq 1$ such that*

$$|\Theta(2^m y)| \geq \tfrac{1}{2}. \tag{42}$$

Proof. Choose the nonterminating binary expansion

$$y = 0.b_1 b_2 b_3 \cdots \quad (b_j \in \{0,1\}).$$

Since y is not dyadic, its digits are not eventually constant: eventual zeros give a terminating expansion, while eventual ones give the alternative expansion of a dyadic rational. Hence there are infinitely many transitions $b_m \neq b_{m+1}$.

Modulo 2, the number $2^m y$ is

$$b_m + 0.b_{m+1} b_{m+2} \cdots .$$

At a transition this lies in $[\frac{1}{2}, \frac{3}{2}]$. By the block formula (7), the centered argument of up then lies in $[-\frac{1}{2}, \frac{1}{2}]$. Since up is even and decreasing on $[0,1]$, (5) gives

$$|\Theta(2^m y)| = \operatorname{up}(\text{centered argument}) \geq \operatorname{up}(\tfrac{1}{2}) = \tfrac{1}{2}.$$

There are infinitely many such m. □

Remark 9.2. The lemma uses only adjacent binary digit changes, not equidistribution or normality. It therefore applies to every non-dyadic point, including highly non-generic automatic, Sturmian, or sparse-digit numbers.

Proposition 9.3 (Exact noncancellation and zero radius at a tie)**.** *Suppose $x \in T_n$, and let k be the unique index such that $B_k = B_{k+1} = H_n(x)$. Put $m_\ell = 6\ell + 4$. Then, for every $\ell \geq 0$,*

$$\begin{aligned} &\left|A_{k,n} B_k^{m_\ell} \Theta(2^{m_\ell} x_k) + A_{k+1,n} B_{k+1}^{m_\ell} \Theta(2^{m_\ell} x_{k+1})\right| \\ &\qquad = A_{k+1,n} H_n(x)^{m_\ell} \operatorname{up}(1/9) \geq \frac{A_{k+1,n}}{2} H_n(x)^{m_\ell}. \end{aligned} \tag{43}$$

Consequently, for all sufficiently large ℓ,

$$\left|G_n^{(m_\ell)}(x)\right| \geq \frac{A_{k+1,n}}{4} Q_{m_\ell} H_n(x)^{m_\ell}, \tag{44}$$

and the Taylor series of G_n at x has radius zero.

Proof. The tie relation and (40) give $a_k = F'(x_k) = 1$. Now

$$F'(\tfrac{1}{4}) = 2F(\tfrac{1}{2}) = 1.$$

Derivative symmetry gives $F'(\frac{3}{4}) = 1$, and strict monotonicity of F' on the two half intervals shows that these are the only interior solutions of $F'(y) = 1$. The exact formal anchors `fabiusReal_quarter` and `fabiusReal_three_quarters` in `InverseQuarterAnchor.lean` give $F(\frac{1}{4}) = \frac{5}{72}$ and $F(\frac{3}{4}) = \frac{67}{72}$. Therefore

$$x_k \in \left\{\tfrac{1}{4}, \tfrac{3}{4}\right\}, \qquad x_{k+1} = F(x_k) \in \left\{\tfrac{5}{72}, \tfrac{67}{72}\right\}.$$

For $m_\ell \geq 4$, both $2^{m_\ell}/4$ and $3 \cdot 2^{m_\ell}/4$ are even integers. The block formula (7) and $\operatorname{up}(\pm 1) = 0$ hence give

$$\Theta(2^{m_\ell} x_k) = 0. \tag{45}$$

Since $64 \equiv 1 \pmod 9$, write $64^\ell = 1 + 9q_\ell$. Directly,

$$2^{m_\ell}\frac{5}{72} = 10q_\ell + 1 + \frac{1}{9},$$
$$2^{m_\ell}\frac{67}{72} = 134q_\ell + 15 - \frac{1}{9}.$$

The displayed integer centers are odd. Thus (7), evenness of up, and monotonicity on $[0, 1]$ show that

$$|\Theta(2^{m_\ell}x_{k+1})| = \mathrm{up}(1/9) \geq \mathrm{up}(1/2) = \frac{1}{2}. \tag{46}$$

Together with $B_k = B_{k+1} = H_n(x)$, equations (45) and (46) prove (43).

Every other spine weight is strictly smaller than $H_n(x)$ by lemma 8.2. After division by $Q_{m_\ell}H_n(x)^{m_\ell}$, their finite sum tends to zero, as does the remainder in (36). Since $A_{k+1,n} > 0$, the reverse triangle inequality gives (44) for all sufficiently large ℓ. Finally, the same estimate as in (51), with $m = m_\ell$, tends to infinity and proves zero Taylor radius. □

10 Proof of the zero-radius theorem

For fixed n, define

$$E_n := (0, 1) \setminus \left(\{\tfrac{1}{2}\} \cup T_n \cup \bigcup_{j=0}^{n-1} G_j^{-1}(\mathcal{D}_2) \right). \tag{47}$$

Every G_j is a homeomorphism, so each preimage $G_j^{-1}(\mathcal{D}_2)$ is countable. The tie set T_n is finite. Thus E_n is co-countable and dense in $(0, 1)$.

Proposition 10.1 (Dominant-spine lower bound)**.** *Let $x \in E_n$, and let j_* be the unique index for which $B_{j_*} = H_n(x)$. There are infinitely many m such that*

$$\left|G_n^{(m)}(x)\right| \geq c(x, n)Q_m H_n(x)^m \tag{48}$$

with a constant $c(x, n) > 0$ independent of m.

Proof. Because $x \in E_n$, the orbit point x_{j_*} is not dyadic. By lemma 9.1, there is an infinite set $M \subset \mathbb{N}$ such that

$$|\Theta(2^m x_{j_*})| \geq \tfrac{1}{2} \qquad (m \in M). \tag{49}$$

For every $j \neq j_*$, put $r_j = B_j/H_n(x) < 1$. Divide (36) by $Q_m H_n(x)^m$. Along M, the dominant summand has absolute value at least $A_{j_*,n}/2$, while the sum of all other spines is bounded by

$$\sum_{j \neq j_*} A_{j,n} r_j^m \longrightarrow 0.$$

The normalized remainder also tends to zero. Therefore, for all sufficiently large $m \in M$, the normalized absolute value is at least $A_{j_*,n}/4$. We may take $c(x, n) = A_{j_*,n}/4 > 0$. □

Proof of theorem 1.2. At every $x \in E_n$, proposition 10.1 and $m! \leq m^m$ give along an infinite subsequence

$$\left(\frac{\left|G_n^{(m)}(x)\right|}{m!}\right)^{1/m} \geq c(x, n)^{1/m} H_n(x) \frac{Q_m^{1/m}}{(m!)^{1/m}} \tag{50}$$

$$\geq c(x, n)^{1/m} H_n(x) \frac{2^{(m+1)/2}}{m} \longrightarrow +\infty. \tag{51}$$

This proves (2). The Cauchy–Hadamard formula then gives zero radius of convergence for the Taylor series at x. □

Proof of theorem 1.1. Suppose G_n were real analytic at some $p \in [0,1]$. Real analyticity is a local open property: there would be a radius $r > 0$ such that G_n is represented by a convergent power series throughout $(p-r, p+r)$, and hence is analytic at every point of that interval. The interval intersects $(0,1)$ in a nonempty open set, even when p is an endpoint. Since E_n is dense, choose $x \in E_n \cap (p-r, p+r)$. This contradicts the zero Taylor radius at x established in theorem 1.2. □

11 Consequences and refinements

11.1 Inverse compositional iterates

Let $I = F^{-1}\colon [0,1] \to [0,1]$. On $(0,1)$ the derivative of F is positive, so F and all G_n are local C^∞ diffeomorphisms.

Corollary 11.1 (Nowhere analyticity of inverse iterates)**.** *For every $n \geq 1$, the inverse iterate*

$$I^{\circ n} = G_n^{-1} \tag{52}$$

is not real analytic at any point of $(0,1)$. Its constant totalization to $\mathbb{R}$ is also nonanalytic at 0 and 1.

Proof. If $I^{\circ n}$ were analytic at an interior point y, its derivative there would be nonzero. The real-analytic inverse-function theorem would make its local inverse G_n analytic at $x = I^{\circ n}(y)$, contradicting theorem 1.1. Analyticity at either endpoint would hold on a one-sided neighborhood and hence at nearby interior points, which has just been ruled out. □

11.2 No local elementary representation

The repository's non-elementarity development proves that every function in its formal elementary class has a dense open analytic locus. The same local obstruction used for F therefore applies to every positive iterate.

Corollary 11.2 (Local non-elementarity)**.** *Fix $n \geq 1$. No elementary function in the repository's real-variable class can agree with $F^{\circ n}$ on a subset of $[0,1]$ having nonempty interior. The same holds for the inverse iterate $I^{\circ n}$ on subsets of $(0,1)$ with nonempty interior.*

Proof. Agreement on an open interval with a function having a dense analytic locus would make the iterate analytic at some point of that interval, contrary to theorem 1.1 and corollary 11.1. □

11.3 A pointwise derivative-growth invariant

For $x \in E_n$, the proof identifies the ordinary exponential correction to the universal dyadic scale:

$$H_n(x) = \max_{0 \leq j < n} (F^{\circ j})'(x). \tag{53}$$

The lower bound along an infinite subsequence is

$$\left| G_n^{(m)}(x) \right| \geq c Q_m H_n(x)^m. \tag{54}$$

The universal factor Q_m does not depend on n. Composition changes only the finite orbit-dependent exponential base and the bounded Thue–Morse amplitude. This explains conceptually why no finite amount of self-composition can restore analyticity.

11.4 A Denjoy–Carleman interpretation

The sequence Q_m satisfies

$$\log Q_m = \frac{\log 2}{2} m^2 + \mathcal{O}(m).$$

It grows faster than every Gevrey sequence $C^m (m!)^s$ of fixed order s, because $\log(m!) = m \log m + \mathcal{O}(m)$. Thus the iterate jets on E_n escape every finite Gevrey class along a subsequence. This is stronger than mere failure of analyticity, although it is a pointwise lower-growth statement rather than a uniform Denjoy–Carleman classification on an interval.

12 Numerical diagnostics

No numerical experiment is used in the proof. The accompanying script `numerical_experiments.py` serves three narrower purposes:

(i) it reconstructs up by an inverse FFT of the truncated sinc product;

(ii) it composes centered Taylor series to evaluate the full Faà di Bruno derivative numerically;

(iii) it compares that derivative with the finite spine sum in (36).

12.1 Sinc-product reconstruction

With the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i \xi t}\, dt$, the Rvachev bump has product

$$\widehat{\mathrm{up}}(\xi) = \prod_{j=0}^{\infty} \mathrm{sinc}\left(\frac{\pi \xi}{2^j}\right), \qquad \mathrm{sinc}(z) = \frac{\sin z}{z}. \tag{55}$$

The script truncates the product, applies an inverse FFT on a period larger than the support, normalizes at $\mathrm{up}(0) = 1$, and uses shape-preserving interpolation. It then recovers $F(x) = \mathrm{up}(x-1)$ for $0 < x < 1$ and the signed blocks (7).

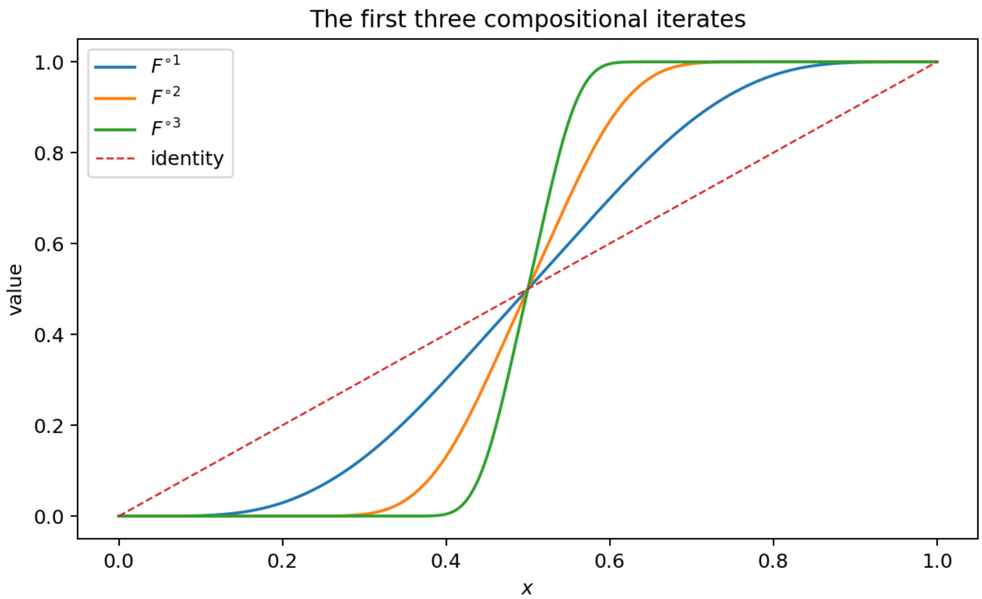

Figure 1: Numerical reconstructions of the first three compositional iterates. The strict drift away from the repelling fixed point $1/2$ and toward the endpoints is visually apparent.

12.2 Taylor-series composition and the spine comparison

The default diagnostic point is

$$x_0 = 0.437123456789, \qquad n = 4.$$

The computed orbit and prefix weights are

$$(x_0, x_1, x_2, x_3) \approx (0.4371234568, 0.3743185303, 0.2522048636, 0.07166875375),$$
$$(B_0, B_1, B_2, B_3) \approx (1, 1.992844087, 3.703442407, 3.768767090).$$

Thus the final orbit level $j = 3$ is the unique dominant spine in this example. Let D_m denote the full numerically composed derivative and S_m the finite spine sum. The following table reports both after division by $Q_m H_4^m$, together with the normalized absolute difference and the m-th root of the Taylor coefficient.

m	$D_m/(Q_m H_4^m)$	$S_m/(Q_m H_4^m)$	$\lvert D_m - S_m\rvert/(Q_m H_4^m)$	$\lvert D_m/m!\rvert^{1/m}$
10	1.049415	0.721343	3.28×10^{-1}	37.84
12	−0.503489	−0.378637	1.25×10^{-1}	60.91
13	0.961480	0.998158	3.67×10^{-2}	84.86
15	−0.384836	−0.384202	6.35×10^{-4}	140.94
16	−1.008935	−1.002630	6.31×10^{-3}	200.75
18	0.432079	0.432434	3.55×10^{-4}	344.83
19	−0.998626	−0.999888	1.26×10^{-3}	486.69
21	0.091358	0.091354	3.90×10^{-6}	793.50
22	0.581063	0.581018	4.50×10^{-5}	1176.18

Table 2: Finite-order diagnostic for the spine asymptotic. Oscillation is expected; the theorem concerns the normalized absolute remainder, not monotone convergence at every order.

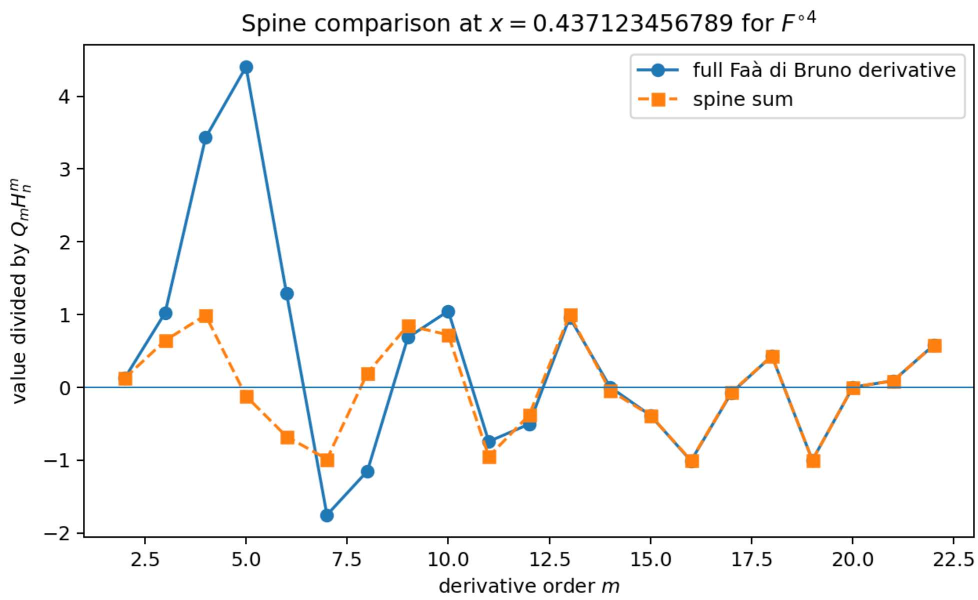

Figure 2: The full derivative and the finite spine sum after the normalization $Q_m H_n^m$.

The script writes all raw values to `numerical_output/spine_diagnostic.csv`, records orbit metadata, and regenerates every figure from command-line

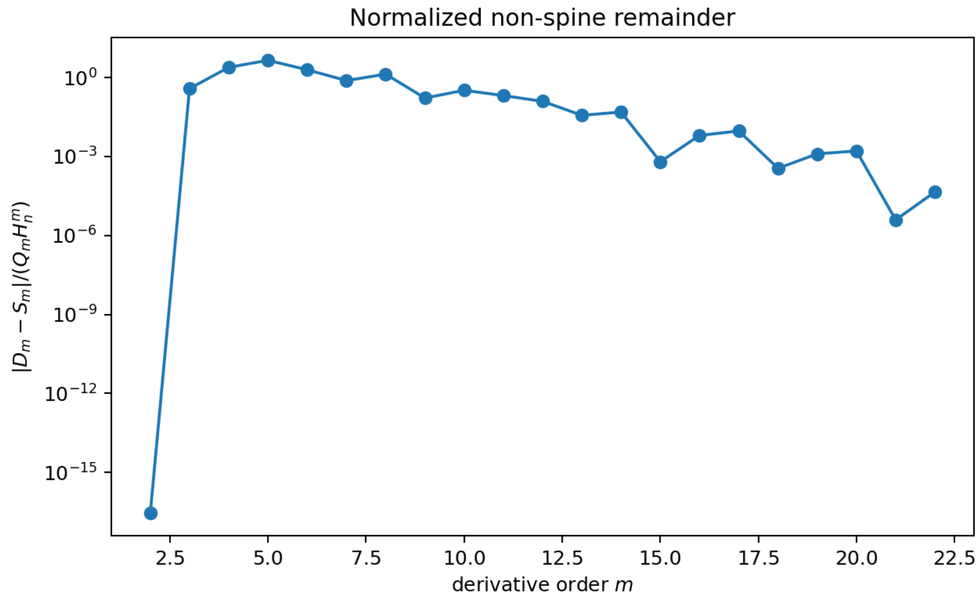

Figure 3: The normalized non-spine remainder. Small-order irregularity is compatible with the asymptotic statement; the high-order values illustrate the defect-driven suppression.

parameters. The code contains detailed comments explaining Fourier normalization, the signed extension, truncated power-series composition, and the spine weights.

13 Connections to the broader Fabius corpus

13.1 Bell polynomials and Bernoulli cumulants

Bell polynomials already occur in the repository through moment and cumulant formulae for dyadic values. Here the same combinatorial species—set partitions—appears in a different role: it organizes the derivatives of a composition. The defect $D(\pi)$ adds a quadratic dyadic energy to a Bell partition. This suggests studying the weighted partition polynomial

$$\mathcal{P}_m(z, w) := \sum_{\pi \in \Pi_m} z^{s(\pi)} w^{D(\pi)}. \tag{56}$$

At $w = 1$ this collapses toward familiar Bell refinements; at $w = 1/2$ it is the exact suppression factor governing the present proof. The Bernoulli-cumulant formulas for up arise from logarithms of dyadic products, while (56) measures the combinatorics of composing the resulting quadratic-scale jets. A q-Bell theory with $q = 1/2$ may unify these two appearances.

13.2 q-binomial and Thue–Morse structure

The exact dyadic values of F are governed by Gaussian-binomial and q-Pochhammer identities, whereas the derivative signs in (12) are Thue–Morse. The present proof shows that under finite self-composition these structures do not spread across all Faà di Bruno trees at leading order. They remain concentrated on a finite set of orbit spines. The leading sign-amplitude process is

$$m \longmapsto \Theta(2^m x_{j_*}), \tag{57}$$

which is a direct symbolic sampling of the dyadic shift. Thus a high-dimensional Bell expansion asymptotically collapses to a one-dimensional Thue–Morse-modulated orbit observable.

13.3 Infinite sinc products

The sinc product (55) supplies the numerical representation used here and explains the exceptional smoothness of up. The nowhere-analyticity mechanism, however, is most transparent in physical space: dilation creates Q_m, block translation creates Thue–Morse signs, and binary transitions prevent the amplitude from becoming too small forever. The Fourier and derivative pictures are therefore complementary. A possible future goal is to derive the spine expansion directly from a multidimensional Fourier integral for $F^{\circ n}$, but nonlinear composition makes such a representation substantially less direct than the set-partition proof.

13.4 Lambert–W endpoint asymptotics

The repository develops small-argument asymptotics of F in which lower real branches of the Lambert–W function encode the saddle point. For a fixed n, the left-half dynamics repeatedly applies an extremely flat map:

$$x \mapsto F(x) \mapsto F(F(x)) \mapsto \cdots . \tag{58}$$

The present derivative theorem is local in the differentiation order m, while the Lambert–W theory is local in $x \downarrow 0$. Their intersection is a two-parameter problem: asymptotics of $G_n^{(m)}(x)$ when both $m \to \infty$ and $x \downarrow 0$. The spine formula predicts phase transitions when the maximizing prefix B_j changes orbit level. Those transition surfaces are natural candidates for nested Lambert–W descriptions.

13.5 Legendre expansions and polynomial representations

The Rvachev function admits rapidly convergent orthogonal-polynomial expansions, and the repository investigates representations of polynomials by shifted up copies. Since G_n becomes increasingly step-like around $1/2$ as n grows, its Legendre coefficients should exhibit a two-scale pattern: smooth compact-kernel decay at fixed n, followed by a singular limit as $n \to \infty$. Theorem 1.1 forbids exponential coefficient decay strong enough to imply analyticity in a Bernstein ellipse touching any real subinterval. Quantifying the exact subexponential decay would connect the present local derivative obstruction to global approximation theory.

14 Conjectures and research directions

The theorem settles nowhere analyticity, but the exceptional countable set left by the proof contains interesting arithmetic structure rather than a genuine gap in the main conclusion.

A tempting strengthening would assert zero Taylor radius at every interior point. That statement is already false for $n = 1$: at a dyadic point, the high-order jet of F eventually vanishes, so the formal Taylor series is a finite polynomial, although it does not represent F on any neighborhood. At any point, exactly one of the following three alternatives holds:

(i) the Taylor series has radius zero;

(ii) only finitely many Taylor coefficients are nonzero, so the series is a finite polynomial and has infinite radius;

(iii) the Taylor series has positive radius and infinitely many nonzero coefficients.

In alternatives (ii) and (iii), theorem 1.1 says that the Taylor series cannot represent G_n on any neighborhood. The requirement of infinitely many nonzero coefficients in (iii) is essential: without it, every series in (ii) would also lie in (iii).

Conjecture 14.1 (Zero-radius or eventually-zero jet)**.** *For fixed n, alternative (iii) is empty. The points in alternative (ii) form a countable orbit-arithmetic set governed by dyadic orbit points.*

This distinction explains why nowhere analyticity does not by itself imply zero Taylor radius. The present theorem proves alternative (i) on a co-countable set and rules out local representation everywhere. Moreover, the tie-cancellation question posed in the submitted version is no longer conjectural: proposition 9.3 proves the exact amplitude on $m = 6\ell+4$, the full-derivative lower bound with constant $A_{k+1,n}/4$, and zero Taylor radius at every tie point. The remaining exceptional Taylor behavior for $n \geq 2$ is therefore associated with dyadic orbit points and their mixed composition jets.

Conjecture 14.2 (Defect-polynomial spectral gap)**.** *For every fixed $z > 0$, the weighted partition sums*

$$\sum_{\substack{\pi\in\Pi_m \\ 1<|\pi|<m}} z^{s(\pi)} 2^{-D(\pi)}$$

admit a full asymptotic expansion whose leading term comes from partitions with one block of size 2 and all remaining blocks singleton, together with the dual family of one block of size $m-1$ and one singleton.

The proof of proposition 5.2 is deliberately coarse. Sharp asymptotics would produce an explicit rate in the spine theorem and might support uniform estimates in x away from orbit tie surfaces.

14.1 Promising formalization route

A Lean formalization can be modular.

1. Define Q_m and prove the algebraic identity (22) for finite set partitions.
2. The block-size-list forms of the pairwise and triangular defect identities, their zero classification, sharp fixed-block minimum, equality profile, and first positive shell are now formalized in `PartitionDefect.lean`. A finite-set-partition wrapper and the quadratic-scale factorization remain open.
3. Replace the asymptotic counting proof initially by a stronger but simpler explicit estimate sufficient for $o(1)$.
4. Use the existing iterated-derivative and bounded-derivative APIs for F.
5. Prove the spine expansion by induction on the fixed iterate count.
6. Reuse the existing strict monotonicity, convexity, order-isomorphism, and Thue–Morse modules to construct E_n and the dominant subsequence.
7. Finish with the existing analytic-at and power-series-radius infrastructure.

Representative exact Lean declarations are listed below; the accompanying repository README gives the complete human-readable API.

```
Fabius.partitionDefect_eq_pairSum_mul_sub_one
Fabius.choose_sum_two_eq_choose_length_add_sum_add_partitionDefect
Fabius.partitionDefect_eq_linear_add_pairwise_excess
Fabius.partitionDefect_fixed_block_bound
Fabius.partitionDefect_fixed_block_eq_iff
Fabius.partitionDefect_eq_zero_iff
Fabius.partitionDefect_eq_firstShell_iff
```

These declarations work more generally for arbitrary positive lists of block sizes. The remaining combinatorial library-building step is the asymptotic weighted enumeration; after that, the analytical induction can follow the existing module style.

14.2 Generalization to other atomic functions

The proof abstracts beyond the classical dyadic Fabius function. Suppose a smooth map f has derivatives

$$f^{(m)}(x) = q^{\binom{m+1}{2}}\Phi(q^m x) \tag{59}$$

for some $q > 1$, with bounded Φ, recurrent nonvanishing along a dense set of q-adic orbits, and monotone orbit slopes that yield a unique dominant prefix. The same defect calculation works with $2^{-D(\pi)}$ replaced by $q^{-D(\pi)}$. This suggests a general theorem for Rvachev-type atomic functions beyond the dyadic case, including multiscale refinable functions with a one-branch derivative law.

15 Summary of the proof mechanism

For reference, the entire argument can be compressed into the following chain.

1. Exact differentiation gives $F^{(m)} = Q_m\Theta \circ (2^m\cdot)$ with $Q_m = 2^{m(m+1)/2}$ and $|\Theta| \le 1$.
2. In Faà di Bruno's set-partition formula, every non-spine partition has a positive quadratic defect $D(\pi)$.
3. The weighted sum of all non-spine partitions is exponentially small, despite the Bell-number multiplicity.
4. After division by Q_m, the derivative $G_n^{(m)}$ is asymptotic to a finite orbit-spine sum with terms $A_{j,n}B_j^m\Theta(2^m x_j)$.
5. Away from a finite tie set, monotone orbit dynamics makes one B_j uniquely maximal.
6. Away from countably many dyadic orbit preimages, binary digit transitions make the dominant amplitude at least $1/2$ infinitely often.
7. Hence $\left|G_n^{(m)}(x)\right| \ge cQ_m H_n^m$ infinitely often on a co-countable dense set.
8. Since $Q_m^{1/m}/(m!)^{1/m} \ge 2^{(m+1)/2}/m \to \infty$, the Taylor radius is zero there.
9. Analyticity at any point would imply analyticity on a neighborhood containing one of those dense zero-radius points, a contradiction.

A Auxiliary details on the partition defect

This appendix records two identities useful for sharpening proposition 5.2.

Proposition A.1 (Exact minimum at fixed block count)**.** *For integers $m \ge 1$ and $1 \le k \le m$,*

$$\min_{\substack{\pi\in\Pi_m \\ |\pi|=k}} D(\pi) = (k-1)(m-k). \tag{60}$$

Equality holds exactly when one block has size $m-k+1$ and every other block is a singleton.

Proof. The indexing family is nonempty: take one block of size $m-k+1$ and $k-1$ singletons. Formula (19) shows that minimizing D is equivalent to maximizing $\sum_i \binom{r_i}{2}$. Strict convexity of $r \mapsto \binom{r}{2}$ transfers one unit from a smaller nonsingleton block to a larger block and increases the sum. Iteration leaves one large block and $k-1$ singletons. Direct substitution gives (60). □

Proposition A.2 (First defect shell). *For $m \geq 3$, the least positive defect is $m-2$. Up to reordering, the only possible block-size profile families are*

$$(m-1,1) \qquad \text{and} \qquad (2,1,\ldots,1). \tag{61}$$

They coincide when $m=3$ and are distinct when $m \geq 4$.

Proof. From (60), the least positive value over $2 \leq k \leq m-1$ is

$$\min_{2\leq k\leq m-1}(k-1)(m-k) = m-2,$$

attained at the endpoint block counts $k=2$ or $k=m-1$; these endpoint values coincide when $m=3$. The equality classification in proposition A.1 gives the displayed profile families. □

This identifies the candidate leading correction to the spine sum. At the near-singleton end, choose one pair and leave all other elements singleton; at the near-giant end, choose the singleton outside a block of size $m-1$. Both carry the same dyadic penalty $2^{-(m-2)}$, but their derivative coefficients and orbit factors are different. A refined expansion should sum these shells before estimating the higher-defect remainder.

B Reproducibility guide

The archive contains the following files.

File	Purpose
`fabius_iterates_nowhere_analytic.tex`	Complete LaTeX source of this report.
`fabius_iterates_nowhere_analytic.pdf`	Rendered report.
`numerical_experiments.py`	Commented Python program for sinc-product reconstruction, Taylor composition, spine evaluation, CSV output, and figures.
`numerical_output/spine_diagnostic.csv`	Raw derivative and spine diagnostics through the selected maximum order.
`numerical_output/numerical_metadata.txt`	Parameters, orbit points, slopes, prefixes, suffixes, and dominant weight.
`figures/*.png`	Figures included in the report.

A typical reproduction command is

```
python numerical_experiments.py \
  --output-dir numerical_output \
  --x0 0.437123456789 \
  --iterate-count 4 \
  --max-order 22
```

The program requires Python 3, NumPy, SciPy, and Matplotlib. It performs consistency checks at $\mathrm{up}(0)$, $\mathrm{up}(1/2)$, $F(0)$, $F(1/2)$, and $F(1)$ before writing output.

C Repository source map

Under the common root `Analysis/FabiusFunction/`, the repository paths most directly connected to the proof are:

- `docs/Fabius_Function_and_Rvachev_Up/` for the primary exposition;
- `GlobalExtension.lean` and `BoundedDerivatives.lean` in `Lean/FabiusFunction/` for the derivative atlas;
- `Convexity.lean`, `Monotonicity.lean`, and the order-isomorphism modules for the orbit dynamics;
- `OriginalPaperSupplement.lean` and `NowhereAnalytic.lean` for the base nonanalyticity proof;
- `PartitionDefect.lean` for the exact finite block-size arithmetic, including the zero profiles, sharp fixed-block minimum, and first positive shell;
- `docs/Non_Elementarity_of_the_Fabius_Function/` for the dense analytic-locus obstruction;
- `docs/semi-formalized-research-frontiers/` for q-series, Bell/Bernoulli, sinc-product, Lambert–W, Legendre, and other frontier connections;
- `docs/archive/` and the frontier README for provenance and consolidation of historical TeX drafts.

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